Conformal Spherical Splines for the Laplace–Beltrami Operator on Genus-Zero Surfaces: Construction, Algorithm, and Experiments

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Abstract

Every smooth closed Riemannian surface of genus zero is isometric to the unit sphere carrying a conformal metric, the round metric multiplied by a positive factor. This paper solves the Poisson equation and the eigenvalue problem of the Laplace–Beltrami operator of such a metric with smooth spherical splines, taking the conformal factor as the only description of the geometry. Because the Dirichlet energy is conformally invariant in two dimensions, the stiffness matrix is the round-sphere matrix for every conformal metric, and the factor enters only the load vector, the mean-value constraint and the weighted mass matrix. A surface embedded in space with a conformal parametrization, a surface of revolution, and a surface given only as a triangle mesh are instances of the same problem that differ in how the factor is obtained; the first two are treated here, and the factor is checked against an area identity before any equation is discretized. The unknown is a spline on a fixed spherical triangulation in broken Bernstein–Bézier form, with C^r smoothness imposed algebraically through edge functionals and the conforming space realized by a null-space matrix computed once and reused for every metric. Each step of the computation is stated with the formulas it needs. Experiments on the round metric, a prolate spheroid, a dumbbell of revolution, an Evans–Fung red-blood-cell profile and a prescribed factor with no embedding show the rates $d+1$ in L^2 and d in energy for spline degree d, the rate $2d$ for eigenvalues with reproduction of the even-order eigenvalues of the round metric up to degree d to roundoff, and, against parametric surface finite elements on the same triangulations, smaller errors at comparable numbers of unknowns for degrees four and six.

Keywords. Laplace–Beltrami operator, Poisson equation on surfaces, spherical splines, conformal parametrization, Bernstein–Bézier methods, genus-zero surfaces
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1 Introduction

Let λ be a smooth function on the unit sphere $\mathbb{S}^2$ and let $\bar{g} = e^{2\lambda} g_S$ be the conformal metric it defines, g_S being the round metric. We write M for the Riemannian surface $(\mathbb{S}^2, \bar{g})$, Δ_M for its Laplace–Beltrami operator and $dA_M = e^{2\lambda}\, dA$ for its area element. For a function f on M with zero mean, the Poisson equation

$$-\Delta_M u = f \quad \text{on } M, \qquad \int_M u \, dA_M = 0, \tag{1}$$

and the eigenvalue problem $-\Delta_M u = \mu u$ are the two model problems for second-order elliptic computation on a surface, and they share their matrices. By the uniformization theorem [29] every smooth closed Riemannian surface of genus zero is isometric to such an M for some λ, so (1) covers the Laplace–Beltrami operator of every smooth closed genus-zero surface, and the isometry is what a conformal parametrization of an embedded surface provides. The question of this paper is practical: given λ, how does one turn either problem into a matrix problem for a smooth high-order spline on the sphere, and what has to be computed at each step? The answer has three parts, and the paper is organized around them.

First, the geometry. The whole of it is the conformal factor $w = e^{2\lambda}$, one positive function on the sphere. How w is obtained depends on the instance: for a surface in $\mathbb{R}^3$ with a conformal parametrization $\Phi\colon \mathbb{S}^2 \to M$, w is the area factor of Φ; for a surface of revolution, w comes from a one-dimensional nonlinear equation, the Mercator matching, and is checked against a closed-form area; a prescribed λ needs no surface at all; and a surface given only as a triangle mesh requires the map to be computed, which is deferred to a separate paper. All four instances feed the same computation (Section 4).

Second, the approximation space. The unknown is a spherical spline of degree d and smoothness r on a fixed spherical triangulation [3, 35]. We write it in broken Bernstein–Bézier form [30]: every spherical triangle keeps its own $\binom{d+2}{2}$ coefficients, C^r smoothness across an edge is a set of linear functionals whose only geometric input is the spherical barycentric coordinates of the opposite vertex, the conforming space is the kernel of the stacked smoothness matrix, and a null-space matrix computed once turns the constrained problem into a symmetric positive definite system. Nothing in this part depends on λ (Section 3).

Third, the equation. Because the Dirichlet energy is conformally invariant in two dimensions, the weak form of (1) in the round metric reads

$$\int_{\mathbb{S}^2} \nabla u \cdot \nabla v \, dA = \int_{\mathbb{S}^2} w\, f\, v \, dA \quad \text{for all } v, \qquad \int_{\mathbb{S}^2} w\, u \, dA = 0, \tag{2}$$

with the round gradient and area element. The stiffness matrix is therefore the round-sphere matrix for every conformal metric, assembled once, and λ enters the load vector and the mean constraint through the single factor w at the quadrature points (Section 5); the eigenvalue problem is the generalized eigenproblem of the same stiffness matrix and the weighted mass matrix. Section 6 lists the computation step by step with the formulas each step evaluates.

The contribution is the formulation in terms of the conformal metric and what it makes computable, rather than a new spline space: the conformal metric on the sphere as the object discretized, with embedded surfaces, surfaces of revolution, prescribed factors and meshed surfaces as instances that differ only in how w is supplied; the geometry carried by one scalar weight without approximation whenever λ is known exactly, so that the discretization itself introduces no geometric consistency error; the metric-independent stiffness matrix and the reusable smoothness and null-space matrices; the eigenvalue formulation with its parity reproduction for the round metric; and a computational comparison with parametric surface finite elements. For the round metric the linear system is that of Baramidze and Lai [10].

The experiments (Section 7) use five instances: the round metric, a prolate spheroid, a strongly necked dumbbell of revolution, the Evans–Fung red-blood-cell profile in physical units, and a prescribed factor with no embedding. On all of them, the spline solution of the Poisson equation converges at the rates $d+1$ in $L^2(M)$ and d in the $H^1(M)$ seminorm expected from the approximation theory of spherical splines [35, Ch. 14], and the eigenvalues converge at the rate $2d$ with reproduction of the even-order eigenvalues of the round metric to roundoff. The

analysis behind the rates, and surfaces given only as triangle meshes, for which λ must itself be computed, are deferred to separate papers; here the emphasis is on what is computed and how.

Throughout, $\mathbb{S}^2$ is the unit sphere with round metric g_S, surface measure dA and tangential gradient ∇; $W^{k,q}$ are Sobolev spaces with norm $\|\cdot\|_{k,q,\Omega}$ and top-order seminorm $|\cdot|_{k,q,\Omega}$; eigenvalues are μ and λ is reserved for the logarithm of the conformal factor; references of the form Theorem 13.30 are to [35].

2 Related work

Spherical splines. The foundational papers are [2, 5, 3, 4], which introduced circular and spherical Bernstein–Bézier polynomials, established dimension and local bases for homogeneous spline spaces, and applied the construction to fitting scattered data on spheres and sphere-like surfaces. The approximation theory and the quasi-interpolation machinery are due to Neamtu and Schumaker [38], [35, Ch. 14]. Spherical splines have been used extensively for data fitting, by minimal energy and least squares [9, 13, 11] and with shape constraints on the Bernstein–Bézier coefficients [12]; such pointwise constraints transfer unchanged to genus-zero surfaces through the parametrization of this paper.

Spherical splines for PDEs, and the sphere-like setting of Baramidze and Lai. The closest prior work is that of Baramidze and Lai [10]: a second-order elliptic equation of Laplace–Beltrami type on the unit sphere is solved by the Galerkin method in $S^r_d(\triangle)$, with existence, uniqueness and convergence proved, the smoothness conditions enforced through the constrained least-squares and augmented Lagrangian machinery of [7], and numerical rates reported; [13] introduces nonhomogeneous spherical splines, restrictions of all trivariate polynomials of degree at most d, so that constants are reproduced for every degree. In the same constrained-coefficient framework, Lai and Lee [33] replace the Galerkin form by collocation, the strong form imposed at interior collocation points and the smoothness conditions imposed as constraints on the coefficients, with the overdetermined constrained system solved by least squares, convergence proved for domains of positive reach, and the approach extended to the three-dimensional Monge–Ampère equation [34]. That realization applies to the transplanted problem of this paper without change, since the strong form $-\Delta_{\mathbb{S}^2}\tilde{u} = w\tilde{f}$ involves only the spherical Laplacian of the Bernstein basis, algebraic in the coefficients by the formulas of Section 3, and the smoothness matrix is the same; we use the Galerkin form because it keeps the stiffness matrix symmetric positive semidefinite and free of the weight, which the eigenproblem needs. Surfaces other than the sphere enter that line of work through the sphere-like construction of [5, 3], a radial graph identified with the sphere by $x \mapsto x/|x|$, which requires M to be star-shaped and puts the surface into both the stiffness and the mass form through a general pulled-back metric. Under the conformal parametrization used here the Dirichlet form is invariant (Lemma 4.1), the stiffness matrix is the round-sphere matrix of [10], and the surface enters only through the scalar weight w; the nonhomogeneous spaces of [13] would lift the parity restriction at the cost of Proposition 5.3.

Finite elements on surfaces. The dominant paradigm is the surface finite element method of Dziuk [23], surveyed in [24] and, for the parametric, trace and narrow-band variants with a full account of the geometric error, in Bonito, Demlow and Nochetto [17]: the surface is approximated by an interpolating polyhedron or an isoparametric lift, and a geometric consistency error enters alongside the Galerkin error. Higher-order parametric elements and their pointwise and a posteriori estimates are due to Demlow [21] and Bonito and Demlow [16], adaptivity on

implicit surfaces to Demlow and Dziuk [22], and the a priori theory for the Laplace–Beltrami eigenvalue problem to Bonito, Demlow and Owen [18]. In all of these high order requires curved elements and lifted quadrature, and C^1 conformity on surfaces is delicate. Here, once a conformal parametrization is known, the geometry is represented through a smooth scalar weight with no further approximation, the discretization introduces no geometric consistency error, C^r smoothness comes from the edge functionals, and the stiffness matrix is independent of the surface. The price is the conformal factor itself, which for a surface given only as a triangle mesh must be computed from an approximately conformal map [27, 32], the subject of a companion paper.

Eigenvalue approximation and spectral geometry. The Galerkin approximation theory of symmetric eigenvalue problems is classical [8, 15]; Hersch's inequality [28] and its extension along the spectrum [31] give a posteriori bounds that every computed genus-zero spectrum must respect.

3 Spherical splines in broken Bernstein–Bézier form

This section sets out the spline machinery in the form in which it is used throughout the paper and in the code. Every spherical triangle keeps its own Bernstein–Bézier coefficients. Smoothness across an edge is a finite set of linear functionals of the coefficients on the two triangles that share the edge. The conforming spline space is the kernel of the matrix obtained by stacking these functionals, and no global basis is ever constructed. This is the Bernstein-constraint representation of [30], specialized to the sphere; the ingredients are those of [35, Ch. 13], and references of the form Theorem 13.30 are to that book. The section closes with the structural features of the spherical theory that shape everything that follows.

3.1 Spherical triangulations

A spherical triangle $T = \langle v_1, v_2, v_3 \rangle$ is the region of $\mathbb{S}^2$ bounded by the great-circle arcs joining $v_1, v_2, v_3 \in \mathbb{S}^2$, assumed to lie in an open hemisphere so that the vertices are linearly independent in $\mathbb{R}^3$. A spherical triangulation $\triangle$ is a finite set of spherical triangles covering $\mathbb{S}^2$, any two meeting in a vertex, an edge, or not at all, with N_V, N_E, N_T vertices, edges and triangles. Every edge belongs to exactly two triangles, and Euler's relation gives

$$N_E = 3N_V - 6, \qquad N_T = 2N_V - 4 \tag{3}$$

(Theorem 13.33). The mesh size $|\triangle|$ is the longest geodesic edge and $\theta_\triangle$ the smallest angle. All experiments use icosahedral refinements: level L has $N_T = 20 \cdot 4^L$ triangles, obtained by splitting every triangle of level $L-1$ at the geodesic midpoints of its edges.

3.2 Spherical barycentric coordinates and Bernstein polynomials

Let $V_T = [\, v_1 \; v_2 \; v_3 \,] \in \mathbb{R}^{3\times 3}$ be the matrix whose columns are the vertices of T. It is invertible because T lies in a hemisphere. The spherical barycentric coordinates of a point $x \in \mathbb{R}^3$ relative to T are

$$b(x) = (b_1, b_2, b_3) = V_T^{-1} x, \qquad \text{so that} \qquad x = b_1 v_1 + b_2 v_2 + b_3 v_3. \tag{4}$$

They are linear, not affine, functions of x: nonnegative on T, with $b_1+b_2+b_3>1$ in the interior of T. The spherical Bernstein basis polynomials of degree d relative to T are

$$B^d_{ijk}(x)=\frac{d!}{i!\,j!\,k!}\,b_1(x)^i b_2(x)^j b_3(x)^k, \qquad i+j+k=d, \tag{5}$$

restricted to $\mathbb{S}^2$. Each is the restriction of a homogeneous trivariate polynomial of degree d, and the $\binom{d+2}{2}$ functions (5) span the space $\mathcal{B}_d$ of spherical polynomials of degree d, which is exactly the restriction to $\mathbb{S}^2$ of all homogeneous polynomials of degree d in $\mathbb{R}^3$ (Section 13.1). Since $\mathcal{B}_d$ does not depend on T, every $p\in\mathcal{B}_d$ has a unique B-form relative to every spherical triangle,

$$p=\sum_{i+j+k=d} c_{ijk}B^d_{ijk}, \tag{6}$$

and we associate the B-coefficient c_{ijk} with the spherical domain point

$$\xi_{ijk}=\frac{iv_1+jv_2+kv_3}{|iv_1+jv_2+kv_3|}, \qquad i+j+k=d. \tag{7}$$

We write $c_T\in\mathbb{R}^{\binom{d+2}{2}}$ for the vector of B-coefficients of p on T, ordered by domain points. Decomposing into spherical harmonics,

$$\mathcal{B}_d=\mathcal{Y}_d\oplus\mathcal{Y}_{d-2}\oplus\cdots \qquad \text{(down to } \mathcal{Y}_0 \text{ or } \mathcal{Y}_1 \text{ by parity)}, \tag{8}$$

where $\mathcal{Y}_\ell$ is the $(2\ell+1)$-dimensional space of spherical harmonics of exact degree ℓ (Theorem 13.22).

For a spherical triangulation $\triangle$ the broken spline space is

$$S_d^{-1}(\triangle)=\{\,s:\mathbb{S}^2\to\mathbb{R}:\ s|_T\in\mathcal{B}_d\ \ \forall T\in\triangle\,\}, \tag{9}$$

with no continuity across edges. A function in $S_d^{-1}(\triangle)$ is determined by the concatenation

$$c=(c_{T_1},\dots,c_{T_{N_T}})\in\mathbb{R}^N, \qquad N=N_T\binom{d+2}{2}, \tag{10}$$

of its B-coefficient vectors on the triangles of $\triangle$. No coefficient is shared between triangles, and a domain point on a common edge carries one coefficient from each of the two triangles that contain it. The spline spaces of interest are

$$S_d^r(\triangle)=\{\,s\in C^r(\mathbb{S}^2):\ s|_T\in\mathcal{B}_d\ \ \forall T\in\triangle\,\}\subset S_d^{-1}(\triangle), \qquad r\ge 0. \tag{11}$$

Every planar Bernstein–Bézier algorithm whose input is a vector of barycentric coordinates applies to (5) verbatim, since the algorithms never use $b_1+b_2+b_3=1$: de Casteljau evaluation, directional differentiation, and subdivision [35, Ch. 13]. Since every global spherical polynomial restricts to each triangle as a member of $\mathcal{B}_d$ and is smooth,

$$\mathcal{B}_d\subset S_d^r(\triangle) \qquad \text{for every } r\ge 0. \tag{12}$$

3.3 Derivatives of the homogeneous extension

Regard B^d_{ijk} as the homogeneous polynomial (5) on $\mathbb{R}^3$. For a direction $z \in \mathbb{R}^3$ let D_z be the directional derivative and $a = (a_1, a_2, a_3) = V_T^{-1} z$ the spherical barycentric coordinates of z. Then

$$D_z B^d_{ijk} = d\big(a_1 B^{d-1}_{i-1,j,k} + a_2 B^{d-1}_{i,j-1,k} + a_3 B^{d-1}_{i,j,k-1}\big), \tag{13}$$

with the convention that a Bernstein polynomial with a negative index is zero (Theorem 13.28). Taking $z = e_1, e_2, e_3$ gives the ambient gradient of the homogeneous extension H_p of $p \in \mathcal{B}_d$ in coefficient form, with $\alpha = (i, j, k)$ and e_m the mth unit multi-index,

$$\frac{\partial H_p}{\partial x_\ell} = d \sum_{|\alpha|=d} c_\alpha \sum_{m=1}^{3} (V_T^{-1})_{m\ell}\, B^{d-1}_{\alpha - e_m}, \qquad \ell = 1, 2, 3. \tag{14}$$

We write G_T for the resulting sparse matrix, the Bernstein gradient matrix of [30] with the rows of V_T^{-1} in the place of the planar barycentric gradients. Euler's identity $x \cdot \nabla H_p(x) = d\, H_p(x)$ gives, for $|x| = 1$, the tangential gradient

$$\nabla_{\mathbb{S}^2} p(x) = (I - xx^\top)\nabla H_p(x) = \nabla H_p(x) - d\, p(x)\, x. \tag{15}$$

The first form is what the assembly evaluates at quadrature points; the second shows that the tangential gradient is algebraic in the coefficients, with ambient components in $\mathcal{B}_{d+1}$ (Theorem 13.15).

3.4 Restriction to an edge

Let $e = \langle v_1, v_2 \rangle$ be the edge of T opposite v_3; it is an arc of the great circle through v_1 and v_2. On e the third coordinate vanishes, $b_3 = 0$, so by (5)

$$p|_e = \sum_{i+j=d} c_{ij0}\, B^d_{ij0}|_e, \qquad B^d_{ij0}(x) = \binom{d}{i} b_1^i b_2^j \ \text{ for } x = b_1 v_1 + b_2 v_2 \in e. \tag{16}$$

The $B^d_{ij0}|_e$ are the linearly independent circular Bernstein polynomials on the arc [2], so two polynomials from the two triangles sharing e agree on e if and only if their coefficients at the $d + 1$ shared domain points agree. In arc length the restriction is a trigonometric polynomial (Remark 13.4), which is why quadrature is unavoidable in assembly.

3.5 Smoothness functionals across an edge

Let $T = \langle v_1, v_2, v_3 \rangle$ and $\widetilde{T} = \langle v_4, v_3, v_2 \rangle$ share the edge $e = \langle v_2, v_3 \rangle$, and let $s \in S_d^{-1}(\triangle)$ have B-coefficients c_{ijk} on T and $\widetilde{c}_{ijk}$ on $\widetilde{T}$, each indexed by the vertex order of its own triangle. Let $(\beta_1, \beta_2, \beta_3) = V_T^{-1} v_4$ be the spherical barycentric coordinates of the fourth vertex relative to T. By Theorem 13.30, s is C^r across e if and only if

$$\widetilde{c}_{njk} = \sum_{\nu+\mu+\kappa=n} c_{\nu,\, k+\mu,\, j+\kappa}\, B^n_{\nu\mu\kappa}(v_4), \qquad j + k = d - n, \quad n = 0, \dots, r, \tag{17}$$

where $B^n_{\nu\mu\kappa}(v_4) = \frac{n!}{\nu!\mu!\kappa!} \beta_1^\nu \beta_2^\mu \beta_3^\kappa$. The first two layers read

$$n = 0: \qquad \widetilde{c}_{0jk} = c_{0kj}, \qquad j + k = d, \tag{18}$$

$$n = 1: \qquad \widetilde{c}_{1jk} = \beta_1 c_{1kj} + \beta_2 c_{0,k+1,j} + \beta_3 c_{0,k,j+1}, \qquad j + k = d - 1. \tag{19}$$

The layer $n = 0$ is the coefficient matching of Section 3.4. Each C^1 condition (19) ties one coefficient of $\widetilde{T}$ in the first row parallel to e to three coefficients of T, the four circled points of Figure 2 in Appendix A; the stencil is the planar one and only the numbers $\beta_1, \beta_2, \beta_3$ change. The layer n involves the coefficients in the rows $0, \dots, n$ parallel to e on both triangles, and the difference of the two sides of (17) is a linear functional of the broken coefficient vector c supported on the two triangles adjacent to e. We call these the smoothness functionals of order r across e and write

$$J_{0,e}^{(r)} \in \mathbb{R}^{m_e \times N}, \qquad m_e = \sum_{n=0}^{r} (d + 1 - n), \tag{20}$$

for the matrix whose rows are these functionals, so that s is C^r across e if and only if $J_{0,e}^{(r)} c = 0$. A row of layer n has at most $\binom{n+2}{2} + 1$ nonzero entries. For $r = 0$ we write $C_{0,e} = J_{0,e}^{(0)}$. The subscript 0 marks the scalar slot of the constraint complex of [30]; only that slot is needed in this paper, since the Laplace–Beltrami problem is scalar and of second order.

The functionals (17) describe C^r continuity on the curved surface exactly: Theorem 13.30 is proved by extending $s|_T$ and $s|_{\widetilde{T}}$ homogeneously to the tetrahedra with apex at the origin, which share the planar face $\langle 0, v_2, v_3 \rangle$, and applying the trivariate C^r conditions across that face. No element map is involved. In the terminology of [30, Sec. 7], where the transfer of C^r functionals to curved patches through element maps is left open, the sphere is the case in which the polynomials live on the surface itself, so the only geometric input is $V_T^{-1} v_4$.

Stacking the blocks $J_{0,e}^{(r)}$ over all N_E edges gives the smoothness matrix of the space,

$$J_0^{(r)} \in \mathbb{R}^{m \times N}, \qquad m = N_E \sum_{n=0}^{r} (d + 1 - n), \qquad S_d^r(\triangle) = \ker J_0^{(r)} \subset S_d^{-1}(\triangle). \tag{21}$$

The implementation stores the edge blocks, so the matrix has $O(N_E)$ rows and nonzeros for fixed d and r. Its rows are linearly dependent, since the conditions around a vertex are related, and this is allowed throughout; the number of dependencies is counted in Appendix A and confirmed on the meshes used.

3.6 Dimension from the rank

The dimension of the spline space is $N - \operatorname{rank} J_0^{(r)}$ whether or not the rows are independent, with the rank delivered by the elimination that builds Z; closed-form counts (Theorem 13.45) are recalled in Appendix A and checked against the computed ranks. For $d < 3r + 2$ the dimension can depend on the positions of the vertices [4], and the rank is then the only reliable count. The entire construction lives on the linear structure of $\mathbb{R}^3$: a general closed surface carries no ambient cone of polynomials, so any extension is a transfer from the sphere, by the radial projection of [5] for star-shaped surfaces or by the conformal parametrization of the next section for every smooth closed genus-zero surface.

4 The conformal metric and its instances

4.1 The conformal metric on the sphere

Let $\lambda \in C^\infty(\mathbb{S}^2)$, $\bar g := e^{2\lambda} g_S$, and $M := (\mathbb{S}^2, \bar g)$. We write $w := e^{2\lambda}$ for the conformal factor. The area element, gradient and Laplace–Beltrami operator of $\bar g$ are

$$dA_M = e^{2\lambda}\, dA, \qquad \nabla_M u = e^{-2\lambda} \nabla u, \qquad \Delta_M u = e^{-2\lambda} \Delta_{\mathbb{S}^2} u, \tag{22}$$

the last being the conformal transformation law in dimension two [29, Ch. 2]. A conformal diffeomorphism of $\mathbb{S}^2$ pulls $\bar g$ back to another conformal metric with the same geometry, so λ is determined by the geometry only up to the noncompact Möbius group [29], and an unnormalized representative can concentrate w arbitrarily. We fix this freedom by Hersch's centering condition [28],

$$\int_{\mathbb{S}^2} x_i \, w \, dA = 0, \qquad i = 1, 2, 3, \tag{23}$$

where x_1, x_2, x_3 are the ambient coordinate functions restricted to $\mathbb{S}^2$; a centered representative always exists [28, 37]. The remaining additive freedom in λ is a global scaling, which we normalize by

$$\operatorname{Area}(M) \;=\; \int_{\mathbb{S}^2} w \, dA \;=\; 4\pi, \tag{24}$$

so that computed spectra are comparable with those of the round metric; the Poisson problem needs neither normalization, and the red blood cell is run in its own units without (24). We write $\Lambda_0 := \max_{\mathbb{S}^2} |\lambda|$ and $E := e^{\Lambda_0}$, so that $E^{-2} \le w \le E^2$; constants C are generic and depend at most on the arguments displayed.

Two forms carry the whole computation. On the round sphere define

$$a(u, v) := \int_{\mathbb{S}^2} \nabla u \cdot \nabla v \, dA, \qquad b_w(u, v) := \int_{\mathbb{S}^2} w \, u \, v \, dA, \qquad u, v \in W^{1,2}(\mathbb{S}^2), \tag{25}$$

and write $\|u\|^2_{L^2_w} := b_w(u, u)$.

Lemma 4.1 (Mass and energy of the conformal metric)**.** *For all $u, v \in W^{1,2}(\mathbb{S}^2)$,*

$$\int_M u \, v \, dA_M = b_w(u, v), \tag{26}$$

$$\int_M \nabla_M u \cdot_{\bar g} \nabla_M v \, dA_M = a(u, v), \tag{27}$$

and $E^{-1} \|u\|_{L^2(\mathbb{S}^2)} \le \|u\|_{L^2_w} \le E \, \|u\|_{L^2(\mathbb{S}^2)}$. In particular $W^{1,2}(M) = W^{1,2}(\mathbb{S}^2)$ with equivalent norms.

Proof. (26) is the first identity of (22). For (27), $\bar g(\nabla_M u, \nabla_M v) = e^{-2\lambda} \nabla u \cdot \nabla v$ pointwise by the second identity; multiplying by $dA_M = e^{2\lambda}\, dA$ the factors cancel. This cancellation is the conformal invariance of the Dirichlet energy, special to dimension two. The norm comparison is $e^{-2\Lambda_0} \le w \le e^{2\Lambda_0}$. □

The approximation power of the spherical spline spaces is Theorem 14.26: if $\triangle$ covers $\mathbb{S}^2$ with $|\operatorname{star}(v)| < 1$ for every vertex and $|\triangle| \le 1/6$, if $d \ge 3r + 2$, and if $0 \le m \le d$ with $m \equiv d \pmod 2$, then for every $g \in W^{m+1,2}(\mathbb{S}^2)$ there is $s \in S^r_d(\triangle)$ with

$$|g - s|_{k,2,\mathbb{S}^2} \;\le\; C\big(d, \theta_\triangle\big)\, |\triangle|^{\,m+1-k}\, |g|_{m+1,2,\mathbb{S}^2}, \qquad 0 \le k \le m, \tag{28}$$

where $\theta_\triangle$ is the smallest angle of $\triangle$ and $|\cdot|_{m+1,2,\mathbb{S}^2}$ is the seminorm of Section 14.5 of [35]. By Lemma 4.1 the same s approximates g in the norms of M: with $m = d$ and $k = 0, 1$,

$$\|g - s\|_{L^2(M)} \;+\; |\triangle|\, |g - s|_{1,2,M} \;\le\; C(d, \theta_\triangle)\, E\, |\triangle|^{\,d+1}\, |g|_{d+1,2,\mathbb{S}^2}, \tag{29}$$

since the energy seminorm of M equals that of the round sphere by (27) and the $L^2(M)$ norm is the weighted norm by (26). The spline spaces themselves need no change: $S^r_d(\triangle)$ is a space of functions on $\mathbb{S}^2$, hence on M, and C^r smoothness, the smoothness matrix $J_0^{(r)}$, the null-space matrix Z, the dimension $N - \operatorname{rank} J_0^{(r)}$, determining sets and local bases are the same for every λ, because none of them involves the metric.

4.2 Instances: how the conformal factor is obtained

An embedded surface with a conformal parametrization. Let $\Sigma \subset \mathbb{R}^3$ be a smooth closed surface of genus zero with its induced metric, and let $\Phi\colon \mathbb{S}^2 \to \Sigma$ be a conformal diffeomorphism, which exists by uniformization [29]. Then $\Phi^* g_\Sigma = e^{2\lambda} g_S$ for one λ, so Φ is an isometry from M onto Σ, and every Laplace–Beltrami problem on Σ is the corresponding problem on M with data $f \circ \Phi$ and solution $u \circ \Phi^{-1}$. The factor is the area magnification of Φ: at $x \in \mathbb{S}^2$ with orthonormal tangent vectors τ_1, τ_2,

$$w(x) \;=\; |\, d\Phi_x(\tau_1) \times d\Phi_x(\tau_2)\,| \;=\; \sigma_1(x)\, \sigma_2(x), \tag{30}$$

where $\sigma_1 \ge \sigma_2 > 0$ are the singular values of $d\Phi_x$, and conformality is $\sigma_1 = \sigma_2$. In colatitude and longitude (Θ, ϕ),

$$w(\Theta, \phi) \;=\; \frac{|\, \partial_\Theta \Phi \times \partial_\phi \Phi\,|}{\sin \Theta}, \tag{31}$$

which a code evaluates when Φ is given with its derivatives; with Φ alone, centered differences with step 10^{-5} reproduced the geometry diagnostics of the experiments below to about ten digits, and the ratio σ_1/σ_2 from the same differences measures the deviation from conformality, assumed here negligible against the discretization error (an approximately conformal map perturbs the metric in both forms; its treatment is deferred). The values of w are needed only at quadrature points.

A surface of revolution. Let Σ be given by a generating curve $(r(\eta), z(\eta))$, $\eta \in [0, \pi]$, rotated about the z-axis, with $r > 0$ on $(0, \pi)$ and the curve meeting the axis regularly at both ends; the spheroid, the dumbbell and the red blood cell are of this form. The metric of Σ in (η, ϕ) is $(r'^2 + z'^2)\, d\eta^2 + r^2\, d\phi^2$, and its Mercator variable is

$$t(\eta) = \ln \tan(\eta/2) + G(\eta), \qquad G'(\eta) = \frac{\sqrt{r'^2 + z'^2}}{r} - \frac{1}{\sin \eta}, \tag{32}$$

where G' extends smoothly to the poles. The round sphere has the same structure with $G \equiv 0$, and matching the two Mercator variables at the equator defines the conformal map $\Theta \mapsto \eta(\Theta)$ through the monotone scalar equation

$$\ln \tan(\eta/2) + \widetilde{G}(\eta) = \ln \tan(\Theta/2), \qquad \widetilde{G} = G - G(\pi/2), \tag{33}$$

solved by a safeguarded Newton iteration, and $\Phi(\Theta, \phi) \;=\; (r(\eta)\cos\phi,\, r(\eta)\sin\phi,\, z(\eta))$. The factor is $w = r(\eta)^2 / \sin^2 \Theta$, which we evaluate in the numerically stable half-angle form

$$w = \Big(\frac{r(\eta)}{\sin \eta}\Big)^2 e^{-2\widetilde{G}(\eta)} \Big(\frac{1 + T^2}{1 + \tau^2}\Big)^2, \qquad \tau = \tan(\eta/2), \quad T = \tan(\Theta/2) = \tau\, e^{\widetilde{G}(\eta)}, \tag{34}$$

which is accurate at the poles. Star-shapedness is not used, so profiles with concavities are admissible. Equatorial symmetry makes the Hersch centering (23) automatic, and the exact area $2\pi \int_0^\pi r\sqrt{r'^2 + z'^2}\, d\eta$ is an independent identity for the computed factor, against which $\int_{\mathbb{S}^2} w\, dA$ under the assembly quadrature is compared as the first check of every run; in the experiments the agreement is at roundoff. For a radial graph $\{R(\psi)\, v(\psi, \phi)\}$ over the sphere the construction is the special case $r = R\sin\psi$, $z = R\cos\psi$, with $G' = (\sqrt{R^2 + R'^2} - R)/(R\sin\psi)$ and prefactor $R(\eta)^2$ in (34); the spheroid is the case with closed-form G.

A prescribed factor. Any smooth λ defines an instance; the computation needs only w at the quadrature points, and Section 7 includes one.

A surface given as a triangle mesh. The map Φ must then be computed, by harmonic energy minimization with Möbius normalization [27] or conformalized mean curvature flow [32], and is only approximately conformal; that instance is the subject of a companion paper.

5 The Poisson equation as a weighted problem on the round sphere

The weak form of (1) is written in the round metric through the two identities of Lemma 4.1 and then in the broken coefficients of Section 3, with conformity imposed through the smoothness matrix. The section closes with the eigenproblem, which uses the same matrices.

5.1 The weak form in the round metric

Let $f \in L^2(M)$ with $\int_M f\, dA_M = 0$. The weak form of (1) is: find $u \in W^{1,2}(M)$ with $\int_M u\, dA_M = 0$ such that

$$\int_M \nabla_M u \cdot_{\bar g} \nabla_M v\, dA_M \;=\; \int_M f\, v\, dA_M \qquad \text{for all } v \in W^{1,2}(M). \tag{35}$$

By (27) on the left and (26) on the right, and since $W^{1,2}(M) = W^{1,2}(\mathbb{S}^2)$, this is the problem on the round sphere: find $u \in W^{1,2}(\mathbb{S}^2)$ with $b_w(u, 1) = 0$ such that

$$a(u, v) \;=\; b_w(f, v) \qquad \text{for all } v \in W^{1,2}(\mathbb{S}^2), \tag{36}$$

with the forms a and b_w of (25); the compatibility condition is $b_w(f, 1) = 0$. Three facts organize what follows. The metric enters only through w, on the right-hand side and in the constraint; a is the Dirichlet form of the round sphere. The problem is well posed: on $\{v : b_w(v, 1) = 0\}$ the form a is coercive, since $a(v, v) \geq \mu_1(M)\, b_w(v, v)$ there by Proposition 5.2 below with $\mu_1(M) > 0$, so Lax–Milgram gives a unique solution, which satisfies (36) for all v because both sides vanish at $v = 1$. The strong form is $-\Delta_{\mathbb{S}^2} u = wf$ by (22); for a manufactured u the product wf in the load vector is $-\Delta_{\mathbb{S}^2} u$ and contains no w, which the first experiment of Section 7 uses to separate the effect of the factor in the load vector from its effect in the constraint and the norms.

5.2 Standing assumptions

Let $\triangle$ be a spherical triangulation with mesh size $|\triangle|$ and smallest angle $\theta_\triangle$, and $S^r_d(\triangle) = \ker J_0^{(r)}$ the spline space of Section 3.5. The standing assumptions are (A1) $\triangle$ covers $\mathbb{S}^2$, $|\mathrm{star}(v)| < 1$ for every vertex and $|\triangle| \leq 1/6$, the hypotheses under which the dimension and approximation theory of spherical splines are available (Theorems 13.43, 13.45, 14.25, 14.26);

(A2) d is even and $d \geq 3r+2$ with $r \geq 0$; (A3) $w = e^{2\lambda}$ with λ smooth, normalized by (23) and, except for surfaces run in physical units, by (24). The parity requirement in (A2) is not technical: $\mathcal{B}_d$ contains the constants if and only if d is even, and by Theorem 13.22

$$\mathcal{B}_d = \mathcal{Y}_0 \oplus \mathcal{Y}_2 \oplus \cdots \oplus \mathcal{Y}_d \qquad (d \text{ even}). \tag{37}$$

Even d puts the constant function, which fixes the mean-value constraint and is the zero-eigenvalue eigenfunction, in the discrete space; for odd d it is not representable even on a single triangle (Theorem 13.23). The inequality $d \geq 3r+2$ is needed only for (28).

5.3 The discrete problem in broken coefficients

Set $S_h := S_d^r(\triangle) \subset W^{1,2}(\mathbb{S}^2)$; $r = 0$ is already conforming, and higher r buys a smoother solution at fewer degrees of freedom. The discrete problem reads: find $u_h \in S_h$ with $b_w(u_h, 1) = 0$ such that

$$a(u_h, v_h) = b_w(f, v_h) \qquad \text{for all } v_h \in S_h. \tag{38}$$

We write it in the broken coefficients of Section 3. On each triangle T the element stiffness matrix K_T, the element weighted mass matrix $M_{w,T}$, and the element load vector F_T, indexed by the multi-indices α, β of degree d, are

$$\begin{aligned} (K_T)_{\alpha\beta} &= \int_T \nabla_{\mathbb{S}^2} B_\alpha^d \cdot \nabla_{\mathbb{S}^2} B_\beta^d \, dA, \qquad (M_{w,T})_{\alpha\beta} = \int_T w \, B_\alpha^d B_\beta^d \, dA, \\ (F_T)_\alpha &= \int_T w \, f \, B_\alpha^d \, dA, \end{aligned} \tag{39}$$

with the tangential gradients given by (15). The broken stiffness matrix $K = \operatorname{diag}_T K_T$ and the broken weighted mass matrix $M_w = \operatorname{diag}_T M_{w,T}$ are block diagonal of size $N \times N$, one block per triangle, the broken load vector $F \in \mathbb{R}^N$ is the concatenation of the F_T, and for $c, y \in \mathbb{R}^N$ representing $s, v \in S_d^{-1}(\triangle)$ one has $y^\top K c = a(s, v)$, $y^\top M_w c = b_w(s, v)$ and $y^\top F = b_w(f, v)$ with the forms extended triangle by triangle. Let $c_1 \in \mathbb{R}^N$ be the broken coefficient vector of the constant function 1. Since $1 \in \mathcal{B}_d$ for even d (Theorem 13.23), its B-form on each triangle is unique, and its entries are not all equal because $b_1 + b_2 + b_3$ is not 1 on the sphere; c_1 is computed once per triangulation and degree by solving the square Bernstein evaluation system at the domain points of each triangle, which is exact up to roundoff. Writing $J := J_0^{(r)}$ for the smoothness matrix, (38) is the constrained linear problem

$$c \in \ker J, \qquad c_1^\top M_w c = 0, \qquad y^\top K c = y^\top F \quad \text{for all } y \in \ker J. \tag{40}$$

The broken stiffness matrix K is the round-sphere Dirichlet matrix, assembled once; M enters only through w in F and $M_w c_1$. This is the constrained Galerkin problem of [30] and the constrained B-form setting of [7]: the broken matrices are never modified, and conformity is a side condition on the coefficient vector; for $w \equiv 1$ it is the discretization of [10]. It can be solved on the same space by a null-space matrix, by the augmented Lagrangian iteration of [7], or as a saddle-point system with Lagrange multipliers [30, Sec. 6]. The experiments use the null-space method [14, Sec. 6], checked against the saddle-point form in Section 7.1.

The null-space method Let $Z \in \mathbb{R}^{N \times n_h}$ have full column rank and range $Z = \ker J$, so that $n_h = \dim S_h$ and every conforming coefficient vector is $c = Z\hat{c}$ for a unique $\hat{c} \in \mathbb{R}^{n_h}$. Substituting into (40) and appending the mean-value constraint with a scalar multiplier ν gives the reduced system

$$\begin{bmatrix} Z^\top K Z & g \\ g^\top & 0 \end{bmatrix} \begin{bmatrix} \hat{c} \\ \nu \end{bmatrix} = \begin{bmatrix} Z^\top F \\ 0 \end{bmatrix}, \qquad g := Z^\top M_w c_1, \qquad c = Z\hat{c}. \tag{41}$$

The reduced stiffness matrix $Z^\top K Z$ is symmetric positive semidefinite with kernel spanned by the reduced coefficients $\hat{c}_1$ of the constant; since $g^\top \hat{c}_1 = \int_{\mathbb{S}^2} w \, dA > 0$, the bordered matrix is nonsingular. The multiplier ν vanishes whenever the discrete compatibility $c_1^\top F = 0$ holds to roundoff, and its size checks the quadrature of the right-hand side. In the language of [35, Ch. 5], the columns of Z extend the coefficients on a determining set to all N broken coefficients.

The construction of Z by sparse rank-revealing elimination of J in a locality order, the saddle-point realization of the same equations, and the reading of residuals in the broken representation (the raw residual $Kc - F$ is a Lagrange multiplier in range $J^\top$ and vanishes only after projection by $Z^\top$) are given in Appendix B.

5.4 What error to expect

The rates observed in Section 7 follow from the approximation estimate (29) by standard arguments, which we state with their hypotheses.

Proposition 5.1. *Let $w \in C^\infty(\mathbb{S}^2)$ with $0 < w_{\min} \le w \le w_{\max}$, let (A1) and (A2) hold, and let $u \in W^{d+1,2}(\mathbb{S}^2)$ solve (1). Then the solution u_h of (38) satisfies*

$$|u - u_h|_{1,2,M} \le C\,|\triangle|^{d}\,|u|_{d+1,2,\mathbb{S}^2}, \qquad \|u - u_h\|_{L^2(M)} \le C\,|\triangle|^{d+1}\,|u|_{d+1,2,\mathbb{S}^2}, \tag{42}$$

with C depending on d, $\theta_\triangle$, $w_{\min}$, $w_{\max}$ and $\|w\|_{C^1(\mathbb{S}^2)}$. For the eigenproblem (45), if μ is an eigenvalue of (43) with eigenspace $E \subset W^{d+1,2}(\mathbb{S}^2)$, then the discrete eigenvalues converging to μ satisfy $0 \le \mu_h - \mu \le C|\triangle|^{2d}$.

Proof. The energy estimate is Céa's lemma [19, Ch. 2] on the w-mean-zero subspace, where a is coercive by Proposition 5.2, with (29) and (27). For the L^2 estimate let $e = u - u_h$ made w-mean-free and let z solve $-\Delta_{\mathbb{S}^2} z = we$, $b_w(z, 1) = 0$; elliptic regularity on the closed manifold $\mathbb{S}^2$ [39, Ch. 5] gives $\|z\|_{W^{2,2}} \le C w_{\max} \|e\|_{L^2}$. Galerkin orthogonality gives $b_w(e, e) = a(z - z_h, e)$ for every $z_h \in S_h$, (28) with $m = 1$ bounds $\inf_{z_h} |z - z_h|_{1,2}$ by $C|\triangle|\|z\|_{W^{2,2}}$, and dividing by $\|e\|_{L^2_w} \ge w_{\min}^{1/2} \|e\|_{L^2}$ and using the energy estimate gives the L^2 rate in the weighted norm, which is the $L^2(M)$ norm by (26). The eigenvalue statement is the Babuška–Osborn estimate for conforming approximation of a symmetric pencil with compact solution operator [8, Sec. 7]: the eigenvalue error is bounded by the square of the best energy approximation of the eigenspace, which (29) bounds by $|\triangle|^{d}$. □

The growth of the constants with the conformal factor and with the position along the spectrum is the subject of a separate paper.

5.5 The Laplace–Beltrami eigenproblem

The eigenvalue problem on M is: find $u \not\equiv 0$ and $\mu \in \mathbb{R}$ with

$$-\Delta_M u = \mu\, u \quad \text{on } M, \tag{43}$$

understood weakly in $W^{1,2}(M)$, and on the round sphere the generalized eigenvalue problem for the pencil (a, b_w): find $u \not\equiv 0$ and $\mu \in \mathbb{R}$ with

$$a(u, v) = \mu\, b_w(u, v) \qquad \text{for all } v \in W^{1,2}(\mathbb{S}^2). \tag{44}$$

Proposition 5.2 (The spectrum as a weighted problem)**.** *(u, μ) is an eigenpair of (43) if and only if it is an eigenpair of (44), and no constant in this correspondence depends on λ.*

Proof. Insert (26) and (27) into the weak form of (43). □

The pencil is symmetric with b_w positive definite, so the spectrum is real, discrete, $0 = \mu_0 < \mu_1 \le \mu_2 \le \cdots \to \infty$ [20, Ch. I], with b_w-orthogonal eigenfunctions and the constant eigenfunction at $\mu_0 = 0$. The discrete eigenvalue problem reads: find $u_h \in S_h \setminus \{0\}$ and $\mu_h \in \mathbb{R}$ with

$$a(u_h, v_h) = \mu_h\, b_w(u_h, v_h) \qquad \text{for all } v_h \in S_h, \tag{45}$$

which in broken coefficients is the constrained pencil $y^\top K c = \mu_h\, y^\top M_w c$ for all $y \in \ker J$, and with the null-space matrix the reduced pencil

$$\left(Z^\top K Z\right) \hat{c} \;=\; \mu_h \left(Z^\top M_w Z\right) \hat{c}, \qquad c = Z\hat{c}, \tag{46}$$

in which $Z^\top M_w Z$ is positive definite. The saddle-point form is the pencil

$$\mathcal{A}x = \mu\, \mathcal{M}x, \qquad \mathcal{A} = \begin{bmatrix} K & J_b^\top \\ J_b & 0 \end{bmatrix}, \qquad \mathcal{M} = \begin{bmatrix} M_w & 0 \\ 0 & 0 \end{bmatrix}; \tag{47}$$

its finite eigenvalues with nonzero primal component are exactly those of (45), and the multiplier directions contribute only infinite eigenvalues [30, Sec. 6], so shift-invert Arnoldi iteration [36] on $(\mathcal{A} - \sigma\mathcal{M})^{-1}\mathcal{M}$ finds the constrained spectrum without any null-space matrix. Three facts follow from the structure. First, since $S_h \subset W^{1,2}(\mathbb{S}^2)$, the Courant–Fischer principle [20, Ch. I] applied over subspaces of S_h gives the one-sided bounds

$$\mu_{i,h} \;\ge\; \mu_i, \qquad i = 0, 1, \ldots, n_h - 1. \tag{48}$$

Second, since d is even, the constants lie in S_h by (12), and $a(1, v_h) = 0 = \mu\, b_w(1, v_h)$ shows that $(1, 0)$ is an exact discrete eigenpair, $\mu_{0,h} = 0 = \mu_0$. Third, the same principle applied to a perturbed weight gives a spectrally uniform stability estimate: if $\widehat{w}$ is the weight actually used and $\delta := \|\widehat{w}/w - 1\|_{L^\infty(\mathbb{S}^2)} < 1$, then $(1 - \delta)b_w \le b_{\widehat{w}} \le (1 + \delta)b_w$ as quadratic forms, and the monotonicity of the Courant–Fischer minimum in the mass form gives

$$\frac{\mu_i}{1 + \delta} \;\le\; \widehat{\mu}_i \;\le\; \frac{\mu_i}{1 - \delta}, \qquad i \ge 0, \tag{49}$$

for the eigenvalues $\widehat{\mu}_i$ of $(a, b_{\widehat{w}})$; the same argument covers quadrature errors in K and M_w. Errors in the weight and the quadrature therefore enter the spectrum as relative errors uniform along the spectrum. Convergence at the rate $|\triangle|^{2d}$ is Proposition 5.1, with no geometric consistency term, in contrast with the surface finite element eigenvalue theory of [18]. One structural statement is specific to the sphere and serves as an implementation test.

Proposition 5.3 (Parity fingerprint)**.** *Let $\lambda \equiv 0$, so that $w \equiv 1$ and (45) is the discrete eigenvalue problem of the round metric, and let d be even. Then for every even ℓ with $0 \le \ell \le d$, every spherical harmonic $Y \in \mathcal{Y}_\ell$ is an exact discrete eigenfunction with exact eigenvalue $\ell(\ell+1)$:*

$$a(Y, v_h) = \ell(\ell+1)\, b_1(Y, v_h) \qquad \text{for all } v_h \in S_h,$$

so the discrete spectrum contains $\ell(\ell+1)$ with multiplicity at least $2\ell+1$, and for $|\triangle|$ small enough the discrete eigenvalue cluster converging to $\ell(\ell+1)$ has total multiplicity exactly $2\ell+1$. The eigenvalues of odd order ℓ converge from above at the rate $|\triangle|^{2d}$.

Proof. By (37) and (12), $\mathcal{Y}_\ell \subset \mathcal{B}_d \subset S_h$ for even $\ell \le d$, and $-\Delta_{\mathbb{S}^2} Y = \ell(\ell+1)Y$ for $Y \in \mathcal{Y}_\ell$ [6, Ch. 2], so $\big(Y, \ell(\ell+1)\big)$ satisfies (45) verbatim; the $2\ell+1$ harmonics of degree ℓ are independent in S_h, which gives the multiplicity bound. By [8, Sec. 7], for $|\triangle|$ small enough exactly $2\ell+1$ discrete eigenvalues, counted with multiplicity, lie in a neighborhood of $\ell(\ell+1)$ that excludes the other eigenvalues of the sphere, and these are the cluster. □

At $d = 6$ the discrete spectrum of the round metric therefore contains 0, 6, 20 and 42 with multiplicities 1, 5, 9 and 13 up to quadrature error, while the clusters at 2, 12 and 30 converge at twelfth order; a quadrature defect perturbs the even clusters, a constraint-rank defect changes the multiplicities, and an assembly error destroys the rate. For every λ, Hersch's inequality $\mu_1(M)\,\mathrm{Area}(M) \le 8\pi$ [28] and its extension $\mu_k(M)\,\mathrm{Area}(M) \le 8\pi k$ [31] bound the eigenvalues from above; since the discrete eigenvalues lie above the exact ones by (48), a normalized discrete eigenvalue exceeding $2k$ signals underresolution or a failure of the normalization (23).

6 The algorithm

Algorithm 1 lists the computation in execution order. The conformal factor is obtained and checked before any equation is discretized, and the smoothness, null-space and stiffness matrices are built once per triangulation and degree and reused for every λ.

The formulas each step evaluates, the Newton iteration and half-angle form for the conformal factor of a surface of revolution, the radial projection of the reference quadrature rule, the quadrature formulas for the load and constraint vectors, the solver diagnostics, and the cost of a new surface or a new right-hand side are given in Appendix C. J, Z and $Z^\top K Z$ are computed once per $(\triangle, d, r)$; a new λ costs one pass over the quadrature points and one factorization of (41); a new right-hand side costs one pass and one triangular solve; the eigenproblem costs one assembly of M_w and one Lanczos run.

7 Numerical experiments

The smoothness matrix J, the null-space matrix Z and the stiffness matrix K are reused unchanged across instances; the metric is supplied through the conformal factor of step 1 of Algorithm 1. Local matrices use a conical Gauss product rule of order 12 per direction on each spherical triangle. The bordered system (41) is solved by a sparse direct solver, and the reduced pencil (46) by shift-inverted Lanczos at $\sigma = -1$. Icosahedral level ℓ has $20 \cdot 4^\ell$ triangles and geodesic mesh sizes 0.628, 0.326, 0.165, 0.083 at levels 1 to 4; the degrees are $(d, r) = (2, 0)$, $(4, 0)$ and $(6, 1)$, the last the smallest even degree admitted by $d \ge 3r + 2$ for C^1 splines.

The instances are the round metric, $\lambda \equiv 0$; three surfaces of revolution, the prolate spheroid $x^2 + y^2 + (z/c)^2 = 1$ with $c = 3/2$, the dumbbell $R(\psi) = 1 - \frac{3}{4}\sin^2\psi$ with two bulbs of radius one

Algorithm 1 Poisson equation for a conformal metric on the sphere by spherical splines

Require: the conformal factor $w = e^{2\lambda}$, as a formula, or as a conformal parametrization Φ or a generating curve from which it is computed; right-hand side f; spherical triangulation $\triangle$; degree d (even) and smoothness r

Ensure: broken coefficient vector c of the spline solution $u_h \approx u$

1: **Geometry.** Build the routine $x \mapsto w(x)$: for a surface of revolution solve (33) for $\eta(\Theta)$ and evaluate (34); for a parametrized surface evaluate (31); for a prescribed λ evaluate $e^{2\lambda}$. Check $\int_{\mathbb{S}^2} w\, dA$ against the exact area when one is known and the Hersch moments (23) against zero.

2: **Spline space (once per $\triangle, d, r$).** Assemble the edge blocks (17) into J; eliminate to reduced row echelon form in the locality order and form Z by (53); compare $N - \operatorname{rank} J$ with the closed-form counts of Appendix A.

3: **Stiffness (once per $\triangle, d$).** Assemble the broken $K = \operatorname{diag}_T K_T$ from (39) with the tangential gradient (15); form and store $Z^\top K Z$.

4: **Weighted data.** At the quadrature points of each triangle evaluate w, f (for an embedded surface, $f \circ \Phi$) and the Bernstein basis; assemble the load vector F and the constraint vector $M_w c_1$ of (39), where c_1 interpolates the constant. Check $c_1^\top F$ against zero.

5: **Solve.** Solve the bordered system (41) by a sparse direct solver; set $c = Z\hat{c}$. Check the multiplier ν against zero and the projected residual $Z^\top(Kc - F)$ against zero.

6: **Output.** Evaluate u_h from c by de Casteljau on any triangle; for an embedded surface the value at $p \in \Sigma$ is $u_h(\Phi^{-1}(p))$. Errors are measured in $L^2(M)$ and $H^1(M)$ through (26) and (27).

joined by a neck of radius $1/4$, and the Evans–Fung red blood cell [25], a biconcave discocyte run in physical units (micrometers), whose factors are computed from the generating curves by (32), with the spheroid's Mercator integral in closed form (50); and a prescribed factor with no embedding, $\lambda = 0.6\,(x_1^2 - x_2^2) - 0.4\,x_3^2$, whose symmetry satisfies (23). The three surfaces of revolution pass the area identity to relative error below 3×10^{-15}; $\operatorname{osc}\lambda$ is 0, 0.535, 2.23, 0.373 and 1.2 respectively, and the dumbbell is the hard case, where $e^{2\operatorname{osc}\lambda} \approx 86$ enters the constants of (29) (Table 7).

7.1 Algebraic checks

The mesh, quadrature, smoothness matrix, null-space matrix and solvers are checked independently of the metric: Euler relations, quadrature area, rank and dependency count of J, reproduction of the constant and of xy, agreement of the null-space and saddle-point realizations to 6×10^{-14}, one-sidedness (48) to 3×10^{-13}, and product-to-moment assembly [1] against direct quadrature to 10^{-15} (Appendix D).

7.2 The Poisson equation

Two tests separate the two places where the metric enters. In Test 1 the exact solution is $u = e^{x_1}x_2 + x_1x_3^3$, not a spherical polynomial of any degree, with $f = -\Delta_M u = -w^{-1}\Delta_{\mathbb{S}^2} u$ computed symbolically; the product wf in the load vector is then $-\Delta_{\mathbb{S}^2} u$, so the discrete solutions for the different instances differ only through the constraint vector $M_w c_1$ and the weighted norm in which the error is measured, and Test 1 isolates the spline approximation from the metric. In

Test 2 the right-hand side is prescribed on the embedded surface, $f(p) = p_1$ made mean-free, so the load vector sees the metric at every quadrature point; its reference is a dense spherical-harmonic Galerkin solve of (36). Errors are measured through (26) and (27) with the assembly quadrature. Table 1 reports Test 1 and Table 2 the degree-4 and 6 rows of Test 2, whose full record is in Appendix E.

Table 1: Poisson problem, Test 1: exact solution $u = \tilde{u} \circ \Phi^{-1}$ with $\tilde{u} = e^{x_1} x_2 + x_1 x_3^3$ on the sphere, right-hand side $f = -\Delta_M u$, icosahedral levels 1 to 4. The $H^1(M)$ seminorm error is the same on all four surfaces, because the load vector $\int_{\mathbb{S}^2} w \tilde{f} B_\alpha \, dA = -\int_{\mathbb{S}^2} (\Delta_{\mathbb{S}^2} \tilde{u}) B_\alpha \, dA$ does not contain w; the $L^2(M)$ error sees w through the constant fixed by the weighted mean and through the weighted norm, and is listed per surface. Rates are against the geodesic mesh size; the $L^2(M)$ rates on the spheroid, dumbbell and red blood cell differ from those of the sphere by at most 0.16 at level 2 and by less than 0.03 at levels 3 and 4. The red blood cell is in micrometers.

			$H^1(M)$, all surfaces		$L^2(M)$, sphere		$L^2(M)$, other surfaces		
(d, r)	lev.	$\dim S_h$	error	rate	error	rate	spheroid	dumbbell	red blood cell
$(2, 0)$	2	642	$4.59\,10^{-2}$	2.04	$1.75\,10^{-3}$	3.04	$2.09\,10^{-3}$	$1.33\,10^{-3}$	$5.56\,10^{-3}$
	3	2562	$1.16\,10^{-2}$	2.02	$2.24\,10^{-4}$	3.02	$2.69\,10^{-4}$	$1.72\,10^{-4}$	$7.11\,10^{-4}$
	4	10242	$2.92\,10^{-3}$	2.01	$2.82\,10^{-5}$	3.02	$3.39\,10^{-5}$	$2.18\,10^{-5}$	$8.96\,10^{-5}$
$(4, 0)$	2	2562	$1.70\,10^{-4}$	4.19	$3.70\,10^{-6}$	5.20	$4.46\,10^{-6}$	$2.77\,10^{-6}$	$1.17\,10^{-5}$
	3	10242	$1.07\,10^{-5}$	4.06	$1.17\,10^{-7}$	5.07	$1.42\,10^{-7}$	$8.84\,10^{-8}$	$3.71\,10^{-7}$
	4	40962	$6.70\,10^{-7}$	4.03	$3.69\,10^{-9}$	5.04	$4.45\,10^{-9}$	$2.78\,10^{-9}$	$1.16\,10^{-8}$
$(6, 1)$	2	3206	$1.40\,10^{-6}$	6.04	$2.57\,10^{-8}$	6.92	$3.25\,10^{-8}$	$2.38\,10^{-8}$	$7.98\,10^{-8}$
	3	12806	$2.49\,10^{-8}$	5.92	$2.48\,10^{-10}$	6.82	$3.14\,10^{-10}$	$2.30\,10^{-10}$	$7.69\,10^{-10}$
	4	51206	$4.22\,10^{-10}$	5.94	$2.25\,10^{-12}$	6.84	$2.82\,10^{-12}$	$1.98\,10^{-12}$	$7.03\,10^{-12}$

In Test 1 the $H^1(M)$ rates over the last refinement are 2.0, 4.0 and 5.9 for $d = 2, 4, 6$, the same to the digits shown across the instances, and the $L^2(M)$ rates are 3.0, 5.0 and 6.8, with the $d = 6$ errors reaching 2×10^{-12} in $L^2(M)$ at 51206 unknowns; the conformal factor enters the $L^2(M)$ errors as a multiplicative constant within the bound E of Lemma 4.1 and nothing else, and the prescribed factor behaves like the embedded surfaces (Appendix E). In Test 2, where the weight is in the load vector, the rates on the sphere, spheroid and red blood cell are again d and $d+1$ to two digits at $d = 2, 4$ and 5.6 to 6.1 in energy at $d = 6$; on the dumbbell the finest $(6, 1)$ row and one $(2, 0)$ row fall short by an amount within the accuracy of the spherical-harmonic reference, as discussed in the appendix. The multiplier ν is below 10^{-13} and the projected residual below 10^{-12} in every row.

Comparison with surface finite elements. Test 1 was repeated on the spheroid and the dumbbell with parametric surface finite elements [23, 21]: Dziuk's polyhedral P_1 method and isoparametric P_2 and P_3 elements on the same icosahedral triangulations pushed to M by Φ, every Lagrange node on the exact surface, so that the geometric error is the smallest the parametric method admits at that degree; errors are measured on the discrete surface with the lifted exact solution. Table 3 reports errors and the wall-clock time of one Python implementation for both methods at the finest level run. At equal degrees of freedom the C^0 spline of degree 2 and the P_2 element are within a factor of two; the spline of degree 4 has an $L^2(M)$ error an order of magnitude below that of P_3 at 10^4 unknowns; and the C^1 spline of degree 6 reaches 3×10^{-10} where P_3 reaches 10^{-6} with twice the unknowns. The finite element errors on the dumbbell are two to three times those on the spheroid at the same level, consistent with the larger geometric approximation error on the more strongly curved surface, whereas the spline energy errors agree

Table 2: Poisson problem, Test 2: right-hand side $f(p) = p_1$ prescribed on the physical surface, so that w enters the load vector at every quadrature point; degrees 4 and 6 (degree 2 and the reference details are in Appendix E). Errors in $L^2(M)$ and the $H^1(M)$ seminorm with rates in parentheses; the red blood cell is in micrometers. The dumbbell reference is converged only to about 10^{-7}, which limits its last rows.

geometry	(d, r)	level	$\dim S_h$	$L^2(M)$ (rate)	$H^1(M)$ (rate)
sphere	$(4, 0)$	2	2562	1.64×10^{-6} (5.21)	7.50×10^{-5} (4.19)
	$(4, 0)$	3	10242	5.20×10^{-8} (5.05)	4.72×10^{-6} (4.05)
	$(6, 1)$	2	3206	9.75×10^{-9} (6.96)	5.63×10^{-7} (6.07)
spheroid	$(4, 0)$	2	2562	2.44×10^{-6} (5.04)	9.69×10^{-5} (4.10)
	$(4, 0)$	3	10242	8.39×10^{-8} (4.94)	6.35×10^{-6} (3.99)
	$(6, 1)$	2	3206	3.50×10^{-8} (6.32)	1.36×10^{-6} (5.60)
dumbbell	$(4, 0)$	2	2562	1.45×10^{-5} (4.30)	3.75×10^{-4} (3.44)
	$(4, 0)$	3	10242	4.05×10^{-7} (5.23)	2.32×10^{-5} (4.07)
	$(6, 1)$	2	3206	8.46×10^{-7} (4.95)	2.79×10^{-5} (4.23)
red blood cell	$(4, 0)$	2	2562	2.27×10^{-4} (5.14)	3.29×10^{-3} (4.13)
	$(4, 0)$	3	10242	7.57×10^{-6} (4.98)	2.15×10^{-4} (3.99)
	$(6, 1)$	2	3206	4.16×10^{-6} (6.63)	6.47×10^{-5} (5.70)

on the two instances to the digits shown, as expected when the stiffness matrix is independent of the metric and the map carries the geometry. In time, the per-instance work of the spline at a given number of unknowns is comparable to P_1 and two to three times P_2, the difference being the order-12 product quadrature of the weighted load vector; the metric-independent setup is amortized over instances. The ratios at the finest configurations are in Table 3.

7.3 The eigenproblem on the round sphere: the parity fingerprint

Table 4 reports the discrete spectra on the round sphere ($w \equiv 1$) against the exact eigenvalues $\ell(\ell + 1)$ with multiplicities $2\ell + 1$, split into the even-ℓ and odd-ℓ clusters of Proposition 5.3. The prediction is realized exactly: the even clusters are reproduced to at worst 2.5×10^{-13}, independent of the mesh, at all three degrees, while the odd clusters converge from above with measured rates 4.02, 8.35, and 11.95 against the expected $2d = 4, 8, 12$ of Section 5.5. At $d = 6$ the exact values $0, 6, 20, 42$ appear with multiplicities $1, 5, 9, 13$. At level 1 the vertex stars have diameter about 1.26, violating $|\mathrm{star}(v)| < 1$ in (A1), and the fingerprint and rates hold regardless.

7.4 The eigenproblem on the curved surfaces

For the spheroid the Mercator matching (33) reads

$$\ln\tan(\theta/2) + G_0(\theta) = \ln\tan(\Theta/2), \qquad G_0(\theta) = \int_{\pi/2}^{\theta} \frac{\sqrt{\cos^2 t + c^2 \sin^2 t} - 1}{\sin t}\, dt, \tag{50}$$

the isometric latitude of the Mercator projection of an ellipsoid, and $\int_{\mathbb{S}^2} w\, dA$ agrees with the closed-form area $2\pi + 2\pi c \arcsin(e)/e$, $e = \sqrt{1 - 1/c^2}$, to relative error 8.4×10^{-16}. None of the three spectra has an elementary closed form, so the reference is a dense Galerkin solve of the same pencil in the real spherical-harmonic basis, in which the zonal weight couples only

Table 3: Conformal spherical splines against parametric surface finite elements (Dziuk P_1, isoparametric P_2, P_3, nodes on the exact surface) for Poisson Test 1 on the spheroid and the dumbbell: finest level run for each degree, errors in $L^2(M)$ and the $H^1(M)$ seminorm with, in parentheses, the convergence rate over the last refinement, and wall-clock seconds. For the splines, "setup is the geometry-independent work (smoothness matrix, null-space matrix, round-sphere stiffness), done once per triangulation and degree and reused for every surface; "surface is the per-surface work (weight at the quadrature points, load and constraint vectors, bordered solve). For the FEM, "setup is the isoparametric mesh and "surface is assembly plus solve. Same machine, single thread, quadrature of order 12 (splines) and 6 (FEM).

geometry	method	dof	$L^2(M)$	$H^1(M)$	setup (s)	surface (s)
spheroid	spline $(2,0)$	10242	$3.39\,10^{-5}$ (3.00)	$2.92\,10^{-3}$ (2.00)	2.3	12.4
	spline $(4,0)$	40962	$4.45\,10^{-9}$ (5.01)	$6.70\,10^{-7}$ (4.01)	3.1	14.4
	spline $(6,1)$	12806	$3.14\,10^{-10}$ (6.79)	$2.49\,10^{-8}$ (5.90)	1.5	5.5
	FEM P_1	10242	$1.44\,10^{-3}$ (2.00)	$9.20\,10^{-2}$ (1.00)	0.8	12.9
	FEM P_2	40962	$4.77\,10^{-6}$ (3.00)	$9.29\,10^{-4}$ (2.00)	1.1	15.1
	FEM P_3	23042	$1.05\,10^{-6}$ (4.02)	$1.53\,10^{-4}$ (3.00)	0.4	4.4
dumbbell	spline $(2,0)$	10242	$2.17\,10^{-5}$ (3.00)	$2.92\,10^{-3}$ (2.00)	2.2	24.2
	spline $(4,0)$	40962	$2.78\,10^{-9}$ (5.01)	$6.70\,10^{-7}$ (4.01)	3.2	25.6
	spline $(6,1)$	12806	$2.30\,10^{-10}$ (6.80)	$2.49\,10^{-8}$ (5.90)	1.4	8.4
	FEM P_1	10242	$1.00\,10^{-3}$ (1.99)	$1.43\,10^{-1}$ (1.00)	0.8	25.4
	FEM P_2	40962	$3.26\,10^{-6}$ (3.06)	$2.55\,10^{-3}$ (1.99)	1.2	27.7
	FEM P_3	23042	$2.14\,10^{-6}$ (4.04)	$6.42\,10^{-4}$ (2.96)	0.5	7.4

Table 4: Round sphere. Maximum eigenvalue errors over the even-ℓ and odd-ℓ clusters (first 16 eigenvalues for $d = 2, 4$; first 50 for $d = 6$), and the measured convergence rate of the odd clusters against the theoretical $2d$. The even clusters are exact up to quadrature and roundoff at every mesh, as Proposition 5.3 predicts.

d	r	level	$\dim S_d^r$	even-ℓ err	odd-ℓ err	rate $(2d)$
2	0	1	162	1.7×10^{-14}	4.94×10^{-2}	
2	0	2	642	3.4×10^{-14}	3.39×10^{-3}	4.09
2	0	3	2562	2.0×10^{-14}	2.18×10^{-4}	4.02
4	0	1	642	1.7×10^{-14}	6.83×10^{-6}	
4	0	2	2562	1.8×10^{-14}	2.88×10^{-8}	8.35
6	1	1	806	1.5×10^{-12}	4.29×10^{-6}	
6	1	2	3206	2.5×10^{-13}	1.71×10^{-9}	11.95

equal azimuthal orders; its self-convergence is given in Table 5. The normalized first eigenvalues $\mu_1 \operatorname{Area}/(4\pi)$ are 1.5514, 0.780 and 1.784, all below the Hersch bound 2; on the red blood cell the equatorial pair $\mu_{1,2} = 0.16715\,\mu\mathrm{m}^{-2}$ lies below the axial mode $\mu_3 = 0.23963$, so in-plane variation is cheaper than top-to-bottom variation on the flat discocyte.

Table 5 gives the rates. On the spheroid ($\operatorname{osc}\lambda = 0.535$) they are 3.98, 7.99 and 10.62 for $d = 2, 4, 6$, the last at an error of 2×10^{-8} where the quadrature of the weight is not yet negligible. On the red blood cell ($\operatorname{osc}\lambda = 0.373$) they are 3.98, 8.02 and 10.65 with terminal errors 1.3×10^{-10} and 3.4×10^{-11} at $d = 4, 6$. On the dumbbell ($\operatorname{osc}\lambda = 2.23$) the coarse errors are orders of magnitude above the spheroid values at identical dimensions and the rates climb toward $2d$ with refinement (3.94 at $d = 2$; 6.31 then 7.00 at $d = 4$; 9.03 at $d = 6$ from one pair of levels), consistent with the expected order and a pre-asymptotic range that widens with $\operatorname{osc}\lambda$. One-sidedness holds to 6×10^{-14} throughout. Figure 1 shows computed eigenfunctions of the three embedded instances.

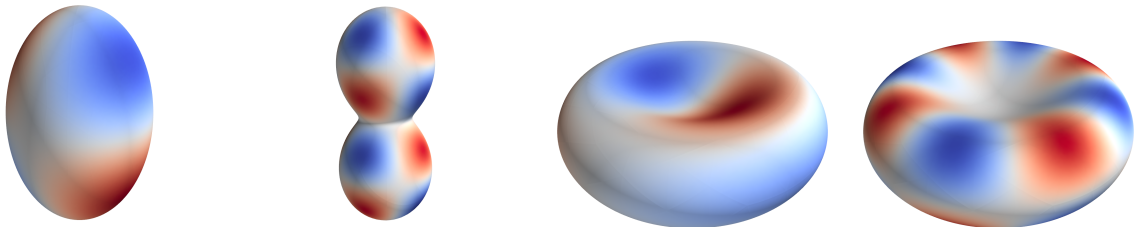


Figure 1: Computed eigenfunctions at $d = 6$, $r = 1$, level 2, drawn on the embedded surfaces. From left: spheroid, seventh eigenfunction, $\mu = 5.38$; dumbbell, $\mu = 57.62$, with azimuthal and axial nodal lines crossing both bulbs; red blood cell, a mode localized in the dimples, $\mu = 1.28\,\mu\mathrm{m}^{-2}$; red blood cell, an azimuthal rim mode, $\mu = 2.66\,\mu\mathrm{m}^{-2}$. Sign and scale are arbitrary.

8 Discussion and conclusions

For a conformal metric on the sphere the area element is the conformal factor times the round one and the Dirichlet energy is the round energy, so the stiffness matrix is the round-sphere matrix for every such metric and the geometry is one scalar at the quadrature points of the load vector, the constraint vector and the weighted mass matrix. By uniformization every smooth closed genus-zero surface is such a metric, and an embedded surface with a conformal parametrization, a surface of revolution, a prescribed factor and a meshed surface are instances that differ only in how the factor is obtained. The factor is checked against an exact area where one exists before any equation is discretized, the C^r spline spaces serve every metric as they stand, and errors in the metric are measured in the round one with no change of constants in energy. The experiments show the rates $d+1$ in $L^2(M)$ and d in energy for the Poisson problem and $2d$ for the eigenvalues on all instances, reproduction of the even-order eigenvalues of the round metric to roundoff, and, on the dumbbell with $\operatorname{osc}\lambda = 2.23$, larger constants and a longer pre-asymptotic range with the limiting orders unchanged.

Relative to surface finite elements [23, 24, 17], the method replaces a sequence of physical meshes by a fixed spherical mesh plus one scalar function. It is restricted to genus zero, needs the conformal factor as input, and a metric with strong conformal concentration maps large

Table 5: Eigenvalues on the curved surfaces (levels 2 and 3; level 1 in Appendix E). Maximum eigenvalue error of the spline pencil against the spherical-harmonic reference (first 16 eigenvalues; first 30 at $d = 6$ on the spheroid) and measured rates against $2d$. Reference self-convergence: 9.4×10^{-13} (spheroid), 8.4×10^{-8} (dumbbell), 2.2×10^{-13} (red blood cell, in $\mu\mathrm{m}^{-2}$).

geometry	d	r	level	$\dim S_d^r$	max error	rate ($2d$)
spheroid	2	0	2	642	4.47×10^{-3}	4.27
	2	0	3	2562	2.94×10^{-4}	3.98
	4	0	2	2562	3.63×10^{-7}	7.99
	6	1	2	3206	1.95×10^{-8}	10.62
dumbbell	2	0	2	642	1.28×10^{0}	2.63
	2	0	3	2562	8.68×10^{-2}	3.94
	4	0	2	2562	4.64×10^{-3}	6.31
	4	0	3	10242	3.88×10^{-5}	7.00
	6	1	2	3206	3.62×10^{-5}	9.03
red blood cell	2	0	2	642	5.84×10^{-4}	4.26
	2	0	3	2562	3.85×10^{-5}	3.98
	4	0	2	2562	3.10×10^{-8}	7.70
	4	0	3	10242	1.29×10^{-10}	8.02
	6	1	2	3206	3.35×10^{-11}	10.65

regions into small spherical patches that a uniform mesh under-resolves; in return, when the factor is known exactly, the geometry is low-dimensional, computed to the accuracy of the area check, and, in the cases run here, at degrees four and six the errors are smaller than those of isoparametric cubic elements at comparable numbers of unknowns. Adaptive refinement driven by the conformal factor is the next step. Two matters are deferred to separate papers: the constants in (42) and in the eigenvalue estimate as functions of the conformal factor and the position along the spectrum, and surfaces given only as triangle meshes, for which the conformal factor must itself be computed.

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Appendix

The appendix collects the implementation details and checks referred to in the main text: the dependency count of the smoothness matrix, the construction of the null-space matrix and the saddle-point realization, the formulas evaluated by each step of the algorithm, the algebraic checks that precede the experiments, and the second Poisson test.

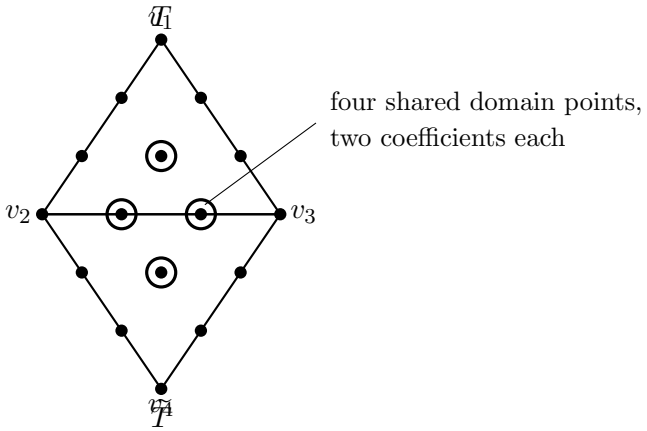


Figure 2: Domain points of two spherical triangles sharing an edge, $d = 3$. Every domain point on the shared edge carries two coefficients, one from T and one from $\widetilde{T}$; the C^0 rows (18) set their differences to zero. One C^1 row (19) ties the four circled coefficients through the spherical barycentric coordinates $(\beta_1, \beta_2, \beta_3)$ of v_4 relative to T. The stencil is the planar one; the geometry of the sphere enters only through the three numbers β_m.

A Dependencies among the smoothness rows

The rows of the smoothness matrix $J_0^{(r)}$ are linearly dependent, since the conditions around a vertex are related. Two counts are elementary. The C^0 rows around a vertex of valence n match the n coefficients at that vertex in a cycle, which has rank $n - 1$, so the C^0 rows carry exactly N_V dependencies. The C^1 rows of the first layer around a vertex match the first-order jets of the n incident polynomials, which by (13) are affine in the vertex coefficient and the two adjacent edge coefficients; the jets form a three-dimensional space, so these n rows have rank $n - 2$ and contribute two further dependencies at every vertex whose incident edges span the tangent plane. Section 7.1 confirms both counts on the meshes used.

The dimension $N - \operatorname{rank} J_0^{(r)}$ of the spline space can be compared with closed-form counts. For $r = 0$ the dimension is the number of domain points of $\triangle$, one per vertex, $d - 1$ per edge and $\binom{d-1}{2}$ per triangle,

$$\dim S_d^0(\triangle) = N_V + (d-1)N_E + \binom{d-1}{2} N_T, \tag{51}$$

for every $d \geq 1$. For $r = 1$ and $d \geq 5$, on a spherical triangulation whose vertices each lie on at least three great circles, so that no vertex is singular, Theorem 13.45 gives

$$\dim S_d^1(\triangle) = \binom{d}{2} N_E - \left(\binom{d+2}{2} - 3\right) N_V + 2\binom{d+2}{2} = \binom{d-1}{2} N_T + 6, \tag{52}$$

the two forms being equal by (3); the constant 6 is three times the Euler characteristic of the sphere. Both identities are checked against the computed ranks in Appendix D.

B The null-space matrix, the saddle-point form, and residuals

Construction of the null-space matrix The null-space matrix is built once per triangulation and degree by sparse Gaussian elimination of the smoothness matrix J to reduced row echelon form, in a column order chosen for locality: the triangles are visited breadth-first from a root triangle, and within a triangle the domain points nearest the edge through which it was entered come first. The elimination is rank revealing [26, Ch. 5], with a pivot tolerance of 10^{-8} relative to the unit entries of J, so dependent rows, which occur around every vertex, and any extra dimensions at singular vertices are handled without case analysis. Let D be the set of pivot columns and F the set of free columns, and let Π be the permutation that orders the columns as (D, F). After elimination,

$$E J \Pi = \begin{bmatrix} I & R \\ 0 & 0 \end{bmatrix}, \qquad Z = \Pi \begin{bmatrix} -R \\ I \end{bmatrix}, \qquad JZ = 0, \qquad |F| = N - \operatorname{rank} J, \tag{53}$$

where E is the product of the row operations and the identity block of Z sits on the free columns. The free coefficients F are a determining set for $S_d^r(\triangle)$: prescribing them determines all pivot coefficients through R. The construction is combinatorial in the mesh and metric only through the coordinates β_m of (19), so Z is the same matrix for every surface M represented on $\triangle$, and in the experiments it is computed once per $(\triangle, d, r)$ and reused for all geometries. Figure 3 shows how the objects fit together.

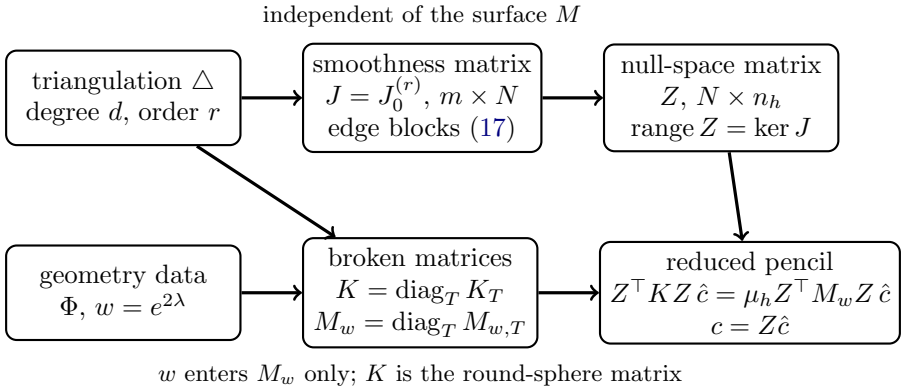


Figure 3: The objects of the discretization and their dependencies. The smoothness matrix J and the null-space matrix Z depend on the triangulation, the degree and the smoothness order only. The broken stiffness matrix K is the round-sphere matrix for every genus-zero surface. The surface enters through the weight w in the broken weighted mass matrix M_w alone.

Rank tolerance and sparsity The dimension of the approximation space depends on the pivot tolerance of the elimination, so its effect is tested. Table 6 (top) repeats the C^1, $d = 6$ construction at level 1 with tolerances 10^{-6} to 10^{-12}: the rank, the dimension and the first eigenvalues of the round sphere agree to all printed digits, so the results are insensitive to the tolerance over this range. Table 6 (bottom) gives the sparsity of Z and of the reduced stiffness matrix $Z^\top K Z$ for levels 1 to 4: the number of nonzeros per column of Z is 6 and the number per row of $Z^\top K Z$ is 63, independent of the level over these four levels, so the reduction generates no growth in fill, and $Z^\top K Z$ has fewer nonzeros than the broken matrix K (4.0×10^6 at level 4). The construction of Z takes under two seconds at level 4 in the Python implementation.

Table 6: C^1 splines of degree 6. Top: rank of J, dimension $N - \operatorname{rank} J$, and the first nonzero eigenvalue and the first eigenvalue of the second cluster of the round sphere at level 1, against the pivot tolerance τ of the elimination. Bottom: size and sparsity of the null-space matrix and of the reduced stiffness matrix by level.

τ	$\operatorname{rank} J$	$\dim S_6^1$	$\mu_{1,h}$	$\mu_{4,h}$
10^{-6}	1434	806	2.0000000008593	6.0000000000000
10^{-8}	1434	806	2.0000000008593	6.0000000000000
10^{-10}	1434	806	2.0000000008593	6.0000000000000
10^{-12}	1434	806	2.0000000008593	6.0000000000000

level	N	n_h	$\operatorname{nnz}(Z)$	$\operatorname{nnz}(Z^\top K Z)$	per col. of Z	per row of $Z^\top K Z$	t_Z (s)
1	2240	806	4744	49766	5.9	61.7	0.03
2	8960	3206	19084	201030	6.0	62.7	0.12
3	35840	12806	76724	805958	6.0	62.9	0.49
4	143360	51206	308324	3225414	6.0	63.0	1.90

The saddle-point form and the broken residual With a full-row-rank J_b spanning the row space of J and Lagrange multipliers λ for the smoothness rows and ν for the mean value, (40) is also the symmetric indefinite system with block matrix $[K, J_b^\top, M_w c_1;\ J_b, 0, 0;\ c_1^\top M_w, 0, 0]$, solved directly without any null-space matrix; the two realizations agree to roundoff (Section 7.1). For the solution c of either, the broken residual $Kc - F + \nu M_w c_1$ is orthogonal to $\ker J$, hence

$$Kc - F + \nu M_w c_1 = -J^\top \lambda \qquad \text{for some } \lambda \in \mathbb{R}^m, \tag{54}$$

because $(\ker J)^\perp = \operatorname{range} J^\top$. The multiplier λ is not zero in general, since single-triangle test functions are not conforming and their interelement flux terms do not cancel. The raw residual is therefore $O(1)$ for the exact discrete solution and vanishes only after projection by $Z^\top$ or subtraction of its component in $\operatorname{range} J^\top$; every residual check in this paper is evaluated in one of these two ways.

C The steps of the algorithm in detail

Step 1: compute and accept the conformal factor The factor is computed by (31) for a general conformal map and by (32)–(34) for a surface of revolution. The equation (33) is solved for η at each quadrature colatitude Θ by a safeguarded Newton iteration with bisection as fallback, which converges to roundoff in a few steps, and the factor is evaluated in the half-angle form (34). Before assembly we record the relative area defect against the exact area, the Hersch residual $\max_i |\int_{\mathbb{S}^2} x_i w \, dA|$, and the oscillation $\operatorname{osc} \lambda = \frac{1}{2}(\ln \max w - \ln \min w)$. The first two decide whether w is accepted; the third predicts where a uniform spherical mesh will underresolve, since a region of M with area fraction θ maps to a spherical region of area fraction of the order θ/w.

Step 3: the stiffness matrix Quadrature on a spherical triangle is the radial projection of a reference plane rule: a flat node u maps to $v = u/\|u\|$ with weight $2A_{\text{flat}}\, h_n \, \|u\|^{-3}$ times the reference weight, where A_{flat} is the area of the flat triangle and h_n its distance from the origin; the factor $2A_{\text{flat}}\, h_n \, \|u\|^{-3}$ is the exact surface Jacobian of the radial projection. The element

matrices K_T use the tangential gradients (14), (15); K contains no w, and $Z^\top K Z$ is formed and stored once per $(\triangle, d, r)$. Both K_T and $M_{w,T}$ can be assembled by the product-to-moment identities of [1], which reproduce direct quadrature to roundoff (Section 7.1).

Step 4: the weighted data On each triangle, at the quadrature nodes v_q with weights W_q, evaluate the Bernstein basis $B^d_\alpha(v_q)$ by de Casteljau, the factor $w(v_q)$ from Step 1, and $f(v_q) = f(\Phi(v_q))$; then

$$\begin{aligned}(F_T)_\alpha &\approx \sum_q W_q\, w(v_q)\, f(v_q)\, B^d_\alpha(v_q),\\ (M_{w,T} c_{1,T})_\alpha &\approx \sum_q W_q\, w(v_q) \Big(\sum_\beta (c_{1,T})_\beta B^d_\beta(v_q)\Big) B^d_\alpha(v_q),\end{aligned} \tag{55}$$

where the inner sum is the constant 1 up to roundoff, c_1 being the exact B-form of the constant. For a manufactured u the product wf is $-\Delta_{\mathbb{S}^2} u$ and the load vector does not see w; for a right-hand side given on M the product $w(v_q) f(\Phi(v_q))$ is where the metric enters. The discrete compatibility $c_1^\top F$ is the quadrature of $\int_M f\, dA_M$ and is checked against zero. For the eigenproblem the full $M_{w,T}$ is assembled by the same rule.

Step 5: solve Form $g = Z^\top M_w c_1$ and $b = Z^\top F$ and solve (41), a symmetric system of size $n_h + 1$ with the pattern of $Z^\top K Z$ plus one dense row and column, by a sparse direct solver; set $c = Z\hat{c}$. Record the multiplier ν and the projected residual $\|Z^\top(Kc - F + \nu M_w c_1)\|/\|Z^\top F\|$, both at roundoff for a direct solve; the raw residual $Kc - F$ is not a diagnostic, by (54). For the eigenproblem, solve (46) by shift-invert Lanczos [36] with a small negative shift; the one-sidedness (48) is monitored in every experiment.

Cost With Z and $Z^\top K Z$ computed once, a new conformal factor costs one pass over the quadrature points for w and f and one factorization of the bordered matrix; a new right-hand side on the same surface costs one pass and one triangular solve; the eigenproblem costs one assembly of M_w and one Lanczos run per surface. Of the checks in Algorithm 1, a failure of the area identity or the Hersch bound is repaired in Step 1, and a failure of reproduction or of the parity fingerprint of Proposition 5.3 in Steps 2 to 5.

D Algebraic checks

Table 7: Geometry diagnostics of the four surfaces, recorded before any solve. The area residual compares the represented area with an independent area calculation.

geometry	map / data	area check	concentration
sphere	exact	rel. $< 3 \times 10^{-15}$	osc $\lambda = 0$
spheroid	analytic	rel. 8.4×10^{-16}	osc $\lambda = 0.535$
dumbbell	revolution	rel. 2.2×10^{-15}	osc $\lambda = 2.23$
RBC	generating curve	at roundoff	osc $\lambda = 0.373$

The Euler relations (3) hold on every mesh and the quadrature area of the sphere is 4π to 3×10^{-15}. At level 1 ($N_V = 42$, $N_E = 120$, $N_T = 80$) the smoothness matrix for $d = 2$,

$r = 0$ has 360 rows and rank 318, and for $d = 6$, $r = 1$ it has 1560 rows and rank 1434; the 42 and $126 = 42 + 84$ dependencies are one, respectively one plus two, per vertex, as counted in Section 3.5. The dimensions $N - \operatorname{rank} J$ are 162, 642, 806 at level 1 and 642, 2562, 3206 at level 2 for S_2^0, S_4^0, S_6^1, in agreement with (51) and (52); the sparse rank agrees with a dense singular value decomposition, and the interpolated constant lies in range Z to 10^{-14}. The constant and the harmonic xy interpolate at the domain points and satisfy every smoothness row to 7×10^{-14}; the constant lies in the kernel of $Z^\top K Z$ to 10^{-14} with mass 4π to 4×10^{-15}. For xy with eigenvalue 6 the broken residual $Kc_{xy} - 6Mc_{xy}$ has relative norm 1.0 in every configuration while its projection by $Z^\top$ is below 6.7×10^{-14}, as (54) states. The reduced pencil (46) and the saddle-point pencil (47), both solved by shift-invert Arnoldi at $\sigma = -1$, give the first sixteen eigenvalues of the round sphere at level 1 to 6×10^{-14} at $d = 2, 4, 6$, so Z spans exactly $\ker J$. The quantity $\min_i(\mu_{i,h} - \mu_i)$ stayed above -3×10^{-13} in every experiment, as (48) requires up to roundoff and reference error. Assembly through the product-to-moment identities of [1] reproduces direct quadrature to 10^{-15} and runs three to five times faster.

E Poisson Test 2 and the convergence plot of Test 1

In Test 2 of Section 7.2 the right-hand side is prescribed on the physical surface, $f(p) = p_1$ made mean-free, so the load vector sees the surface at every quadrature point. Its reference is a dense Galerkin solve of the transplanted problem in the spherical-harmonic sector $\overline{P}_\ell^1(\cos\Theta)\cos\phi$, $\ell \le L = 80$, to which the zonal weight and the azimuthal order of f confine the problem; its coefficients are converged to 10^{-12} on the sphere, spheroid and red blood cell, and to about 10^{-7} on the dumbbell, whose concentrated weight needs many more harmonics. Table 8 reports the results. On the sphere, spheroid and red blood cell the rates are d and $d + 1$ to two digits at $d = 2, 4$ and 5.6 to 6.1 in energy at $d = 6$. On the dumbbell the rates at $(2, 0)$ level 3 (2.15, 1.64) and $(6, 1)$ level 2 (4.95, 4.23) fall short; the errors there, 7.7×10^{-5} and 8.5×10^{-7}, are within a factor of a few of the reference accuracy, so the deficit is attributable to the reference, and the $(4, 0)$ sequence on the same surface shows 5.23 and 4.07. Figure 4 plots Test 1 against the mesh size.

Table 9 gives Test 1 for the prescribed factor $\lambda = 0.6\,(x_1^2 - x_2^2) - 0.4\,x_3^2$, an instance with no embedded surface: the metric is defined by λ alone, the Hersch condition holds by symmetry, and the same computation applies with $w = e^{2\lambda}$ evaluated at the quadrature points. The rates are those of the embedded instances.

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Table 8: Poisson problem, Test 2: right-hand side $f(p) = p_1$, the first Cartesian coordinate of the point $p \in M$, so that w enters the load vector; reference solution from a spherical-harmonic Galerkin solve with $L = 80$, converged to 10^{-12} in the coefficients on the sphere, spheroid and red blood cell and to about 10^{-7} on the dumbbell. Errors in $L^2(M)$ and in the $H^1(M)$ seminorm; rates against the geodesic mesh size. The red blood cell is in physical units (micrometers), so its errors carry those units.

geometry	(d, r)	level	$\dim S_h$	$L^2(M)$ error	rate	$H^1(M)$ error	rate
sphere	$(2, 0)$	1	162	4.04×10^{-3}		4.88×10^{-2}	
	$(2, 0)$	2	642	5.06×10^{-4}	3.17	1.23×10^{-2}	2.10
	$(2, 0)$	3	2562	6.34×10^{-5}	3.04	3.10×10^{-3}	2.02
	$(4, 0)$	1	642	4.97×10^{-5}		1.17×10^{-3}	
	$(4, 0)$	2	2562	1.64×10^{-6}	5.21	7.50×10^{-5}	4.19
	$(4, 0)$	3	10242	5.20×10^{-8}	5.05	4.72×10^{-6}	4.05
	$(6, 1)$	1	806	9.30×10^{-7}		3.00×10^{-5}	
	$(6, 1)$	2	3206	9.75×10^{-9}	6.96	5.63×10^{-7}	6.07
spheroid	$(2, 0)$	1	162	6.36×10^{-3}		6.52×10^{-2}	
	$(2, 0)$	2	642	7.87×10^{-4}	3.19	1.66×10^{-2}	2.09
	$(2, 0)$	3	2562	9.90×10^{-5}	3.04	4.17×10^{-3}	2.02
	$(4, 0)$	1	642	6.65×10^{-5}		1.43×10^{-3}	
	$(4, 0)$	2	2562	2.44×10^{-6}	5.04	9.69×10^{-5}	4.10
	$(4, 0)$	3	10242	8.39×10^{-8}	4.94	6.35×10^{-6}	3.99
	$(6, 1)$	1	806	2.20×10^{-6}		5.33×10^{-5}	
	$(6, 1)$	2	3206	3.50×10^{-8}	6.32	1.36×10^{-6}	5.60
dumbbell	$(2, 0)$	1	162	3.70×10^{-3}		3.00×10^{-2}	
	$(2, 0)$	2	642	3.33×10^{-4}	3.68	7.63×10^{-3}	2.09
	$(2, 0)$	3	2562	7.68×10^{-5}	2.15	2.49×10^{-3}	1.64
	$(4, 0)$	1	642	2.41×10^{-4}		3.56×10^{-3}	
	$(4, 0)$	2	2562	1.45×10^{-5}	4.30	3.75×10^{-4}	3.44
	$(4, 0)$	3	10242	4.05×10^{-7}	5.23	2.32×10^{-5}	4.07
	$(6, 1)$	1	806	2.17×10^{-5}		4.44×10^{-4}	
	$(6, 1)$	2	3206	8.46×10^{-7}	4.95	2.79×10^{-5}	4.23
red blood cell	$(2, 0)$	1	162	5.47×10^{-1}		2.11×10^{0}	
	$(2, 0)$	2	642	6.80×10^{-2}	3.18	5.38×10^{-1}	2.09
	$(2, 0)$	3	2562	8.53×10^{-3}	3.04	1.35×10^{-1}	2.02
	$(4, 0)$	1	642	6.55×10^{-3}		4.91×10^{-2}	
	$(4, 0)$	2	2562	2.27×10^{-4}	5.14	3.29×10^{-3}	4.13
	$(4, 0)$	3	10242	7.57×10^{-6}	4.98	2.15×10^{-4}	3.99
	$(6, 1)$	1	806	3.19×10^{-4}		2.71×10^{-3}	
	$(6, 1)$	2	3206	4.16×10^{-6}	6.63	6.47×10^{-5}	5.70

Table 9: Poisson Test 1 for the prescribed factor $\lambda = 0.6\,(x_1^2 - x_2^2) - 0.4\,x_3^2$ with no embedding ($\operatorname{osc}\lambda = 1.2$). Errors in $L^2(M)$ and the $H^1(M)$ seminorm with rates against the geodesic mesh size.

(d, r)	level	$\dim S_h$	$L^2(M)$	rate	$H^1(M)$	rate
$(2, 0)$	1	162	1.26×10^{-2}		1.75×10^{-1}	
	2	642	1.72×10^{-3}	3.04	4.59×10^{-2}	2.04
	3	2562	2.20×10^{-4}	3.01	1.16×10^{-2}	2.01
	4	10242	2.77×10^{-5}	3.00	2.92×10^{-3}	2.00
$(4, 0)$	1	642	1.12×10^{-4}		2.64×10^{-3}	
	2	2562	3.68×10^{-6}	5.22	1.70×10^{-4}	4.19
	3	10242	1.17×10^{-7}	5.05	1.07×10^{-5}	4.05
	4	40962	3.66×10^{-9}	5.01	6.70×10^{-7}	4.01
$(6, 1)$	1	806	2.32×10^{-6}		7.34×10^{-5}	
	2	3206	2.49×10^{-8}	6.92	1.40×10^{-6}	6.05
	3	12806	2.42×10^{-10}	6.79	2.49×10^{-8}	5.90
	4	51206	2.19×10^{-12}	6.81	4.22×10^{-10}	5.91

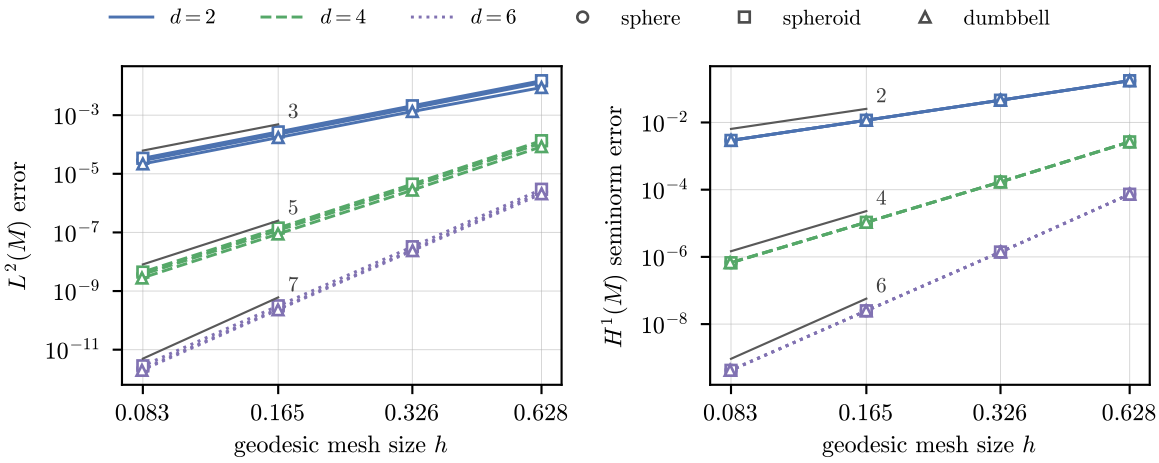


Figure 4: Poisson problem, Test 1, icosahedral levels 1 to 4. Left: error in $L^2(M)$; right: error in the $H^1(M)$ seminorm, against the geodesic mesh size, for $d = 2, 4, 6$ (line style) on the sphere, the spheroid and the dumbbell (marker). The short gray segments have the slopes $d + 1$ (left) and d (right). The three geometries coincide in energy because the load vector of Test 1 does not contain the conformal factor, and are within a factor E of each other in $L^2(M)$. The red blood cell, in micrometers, is omitted from the plot.

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